BAD702 – Statistical Machine Learning for Data Science

Module 3: Statistical Experiments and Significance Testing • VTU Dec 2025/Jan 2026 paper • Answers based only on the textbook

Q.5 a. Describe the concept of P – value in hypothesis testing. Why it is important and how is it interpreted? [10 Marks]

1. Introduction – hypothesis tests

Hypothesis tests (also called **significance tests**) are used in the traditional statistical analysis of published research. Their purpose is to help us learn whether **random chance** might be responsible for an observed effect. The classical inference pipeline is: **formulate hypothesis → design experiment → collect data → inference/conclusions**. Statistical hypothesis testing was invented to protect researchers from being fooled by random chance.

- **Null hypothesis (H_0):** the hypothesis that chance is to blame – the treatments are equivalent and any difference is due to chance.
- **Alternative hypothesis (H_1):** the counterpoint to the null, i.e. what we hope to prove.
- The hypothesis test assumes H_0 is true, builds a "null model" (a probability model) and checks whether the observed effect is a reasonable outcome of that model.

2. Concept of the p-value

Definition: Given a chance model that embodies the null hypothesis, the **p-value is the probability of obtaining results as unusual or extreme as the observed results**. In other words, it is the frequency with which the chance model produces a result more extreme than the observed result.

p-value = P(result as extreme as observed | H_0 is true)

What people want it to mean: "the probability that the result is due to chance". **What it actually is:** the probability that, *given a chance model*, results as extreme as the observed results could occur. The difference is subtle but real – a significant p-value does not carry you as far along the road to "proof" as it seems to promise.

3. How the p-value is estimated (permutation test)

1. Combine the results of both groups into one data set (this represents H_0).
2. Shuffle and randomly divide it into resamples of the original group sizes; record the test statistic (e.g. difference in conversion rate).
3. Repeat many times (R = 1,000) to get the permutation distribution.
4. p-value = proportion of times the permuted difference is equal to or greater than the observed difference.

Textbook example (price test): price A converted 200 of 23,739 and price B converted 182 of 22,588. Observed difference = 0.0368 percentage points. In R: `mean(perm_diffs > obs_pct_diff)` gave **p = 0.308**, i.e. a result this extreme or more would occur by chance over 30% of the time. The normal approximation (`prop.test`) gave p = 0.3498, close to the permutation value.

4. Alpha – the threshold

Statisticians frown on leaving it to the researcher's discretion to decide whether a result is "too unusual". A threshold called **alpha** is specified **in advance**, e.g. "more extreme than 5% of the chance (null hypothesis) results". Typical alpha levels are **5% and 1%**. The chosen level is an arbitrary decision. Alpha is the probability threshold of "unusualness" that chance results must surpass for actual outcomes to be deemed statistically significant.

5. Why the p-value is important

- **Protects against random chance:** the human mind underestimates natural random behaviour (e.g. people inventing coin-flip sequences avoid long runs), and treats a chance pattern as real. The p-value gives an objective yardstick.
- **Decision aid in A/B tests:** tells whether the difference between treatments A and B lies within the range of chance variation.
- **Controls Type 1 error:** significance tests are structured to minimize Type 1 errors (mistakenly concluding an effect is real when it is due to chance).
- **Use in data science:** a useful metric to know whether a model result that looks interesting is within normal chance variability; p-values are sometimes used as intermediate inputs, e.g. to include or exclude a feature in a model. As a decision tool it should not be controlling but only another point of information.

6. Interpretation of the p-value

p-value	Interpretation
Small ($p < \alpha$, e.g. < 0.05)	Observed result is unlikely under H_0 → reject H_0; the result is **statistically significant** (beyond chance variation).
Large ($p > \alpha$), e.g. 0.308	Observed result is well within the range of chance variation → H_0 not rejected; "**effect not proven**".

- A non-significant p-value is not a Type 2 error as such; it is a judgement that the sample size may be too small to detect the effect. A larger sample could yield a smaller p-value.
- A p-value is **not** "the probability that the result happened by chance". It is the probability of a result this extreme given a chance model; the judgement about the chance model itself does not carry a probability.
- **Practical significance:** even if statistically significant, a result may not be practically important. Large samples can make small, non-meaningful effects big enough to rule out chance, but ruling out chance does not make an unimportant result important.

7. ASA statement on p-values (March 2016) – six principles

1. P-values can indicate how incompatible the data are with a specified statistical model.
2. P-values do not measure the probability that the studied hypothesis is true, or that the data were produced by random chance alone.
3. Scientific conclusions and business or policy decisions should not be based only on whether a p-value passes a specific threshold.
4. Proper inference requires full reporting and transparency.
5. A p-value, or statistical significance, does not measure the size of an effect or the importance of a result.
6. By itself, a p-value does not provide a good measure of evidence regarding a model or hypothesis.

8. p-value controversy

One psychology journal even "banned" p-values because decisions based only on them produced poor research: researchers root around in the data until some hypothesis gives a significant p-value (data "dredging").

9. Type 1 and Type 2 errors

- **Type 1 error:** concluding an effect is real when it is due to chance.
- **Type 2 error:** concluding an effect is due to chance when it is real.

Conclusion

The p-value is the probability, under the null (chance) model, of a result as extreme as observed. A small p-value (below alpha) rules out chance; a large one means "effect not proven". Read it with effect size and practical significance.

Q.5 b. Write the step by step procedure for performing a two – sample t – test? [10 Marks]

1. Introduction

There are numerous types of significance tests depending on whether the data are count data or measured data, how many samples there are and what is being measured. A very common one is the **t-test**, named after **Student's t-distribution**, originally developed by **W. S. Gosset**. It is used for the very common **two-sample comparison (A/B test) in which the data is numeric**.

In the 1920s–30s it was not feasible to shuffle data thousands of times for a resampling test. Statisticians found that the t-test, based on the t-distribution, was a good approximation to the permutation (shuffled) distribution. To use the t-distribution without regard to scale, a **standardized form of the test statistic** (the t-statistic) must be used.

Key term	Meaning
Test statistic	A metric for the difference or effect of interest.
t-statistic	A standardized version of common test statistics such as means.
t-distribution	A reference distribution (derived from the null hypothesis) to which the observed t-statistic is compared.

2. Step-by-step procedure

1. **State the hypotheses.** H_0: there is no difference between the means of group A and group B (any difference is chance). H_1: the means differ (two-tail), or a directional alternative such as "mean of A is less than mean of B" (one-tail). Together they must account for all possibilities.
2. **Fix alpha and the type of test in advance** – typically 5% or 1%; choose a one-tail test if the new option must prove itself better than the default, otherwise a two-tail test. Also decide the single **test statistic/metric** (e.g. mean session time) beforehand to avoid researcher bias.
3. **Design the experiment and collect data.** Randomly assign subjects to the two groups (control and treatment). Example: 36 web sessions – 21 for page A and 15 for page B.
4. **Explore the data.** Compare the groups with side-by-side boxplots and compute the group means. Example: mean A = 126.33 s, mean B = 162.00 s; difference = 35.67 s.
5. **Compute the t-statistic.** The difference between the group means is standardized (scaled by its variability) so it can be compared with the t-distribution. The formulas are built into software, so the function is called (R: `t.test`; Python: `scipy.stats.ttest_ind`). The degrees of freedom are also obtained.
6. **Find the p-value** by comparing the observed t-statistic with the t-distribution (the reference distribution under H_0).
7. **Decide.** If $p < \alpha$, reject H_0 (statistically significant); if $p > \alpha$, H_0 is not rejected – the difference lies within chance variation.
8. **Conclude** in the context of the problem and, if possible, cross-check with a permutation test.

3. Textbook example – R code and output

```
> t.test(Time ~ Page, data=session_times, alternative='less')

        Welch Two Sample t-test

data:  Time by Page
t = -1.0983, df = 27.693, p-value = 0.1408
alternative hypothesis: true difference in means is less than 0
95 percent confidence interval:
     -Inf 19.59674
sample estimates:
mean in group Page A  mean in group Page B
            126.3333              162.0000
```

Python:

```
res = stats.ttest_ind(session_times[session_times.Page == 'Page A'].Time,
                      session_times[session_times.Page == 'Page B'].Time,
                      equal_var=False)
print(f'p-value for single sided test: {res.pvalue / 2:.4f}')
```

4. Reading the result

- Alternative hypothesis: the session-time mean for page A is **less than** that for page B (one-tail test).
- t = −1.0983 with df = 27.693 and **p = 0.1408**. At α = 0.05, p > α, so H_0 is **not rejected**.
- Conclusion: although page B's mean is 35.67 s longer, the difference is within the range of chance variation – **not statistically significant**.
- The p-value 0.1408 is fairly close to the permutation test p-values of 0.121 and 0.126 for the same data, showing that the t-test approximates the permutation test.

5. Notes on the procedure

- **Welch version:** the output is a Welch two-sample t-test (`equal_var=False` in Python), which does not require the two groups to have equal variance and works with unequal sample sizes (21 and 15).
- **One-tail p-value in Python:** software typically gives a two-tail p-value, so the value is divided by 2 for the single-sided test (`res.pvalue / 2`). R and scipy default to two-tail output.
- **Degrees of freedom** are part of the standardization so that the t-statistic matches the correct reference distribution.
- The test statistic must be selected **before** the experiment – selecting it afterwards opens the door to researcher bias.

6. t-test versus resampling (permutation) test

Point	t-test	Permutation (resampling) test
Reference distribution	Standard t-distribution (formula based)	Created from the data itself by shuffling
Scale of data	Needs a standardized statistic	Scale does not matter; statistic used as is
Assumptions	Formula variations; data numeric	No normality assumption; numeric or binary; sample sizes may differ
Computation	Possible in the 1920s–30s (before computers)	Needs thousands of shuffles (computers)

- Before computers, resampling was impractical, so statisticians used standard reference distributions; the t-statistic is one such widely used standardized statistic.
- In a resampling mode the solution reflects the observed data and the hypothesis, without worrying about whether the data is numeric or binary, balanced or unbalanced sample sizes, or variances.

Conclusion

A two-sample t-test compares two group means by standardizing their difference into a t-statistic, referring it to the t-distribution with the proper degrees of freedom, obtaining the p-value and comparing it with the pre-chosen alpha to reject or retain H_0. In the page A vs page B example, p = 0.1408 > 0.05, so the difference in session time is not significant.

Q.6 a. What is multiple testing problem? How does it affect statistical inference and how is it corrected? [10 Marks]

1. Introduction – the multiple testing problem

There is a saying in statistics: **"Torture the data long enough, and it will confess."** If you look at the data through enough different perspectives and ask enough questions, you almost invariably will find a statistically significant effect. This is the **multiple testing (multiplicity) problem**: the more tests, variables, models or questions you try on the same data, the greater the probability that something will emerge as "significant" just by chance.

Key term	Meaning
Type 1 error	Mistakenly concluding that an effect is statistically significant.
False discovery rate	Across multiple tests, the rate of making a Type 1 error.
Alpha inflation	The phenomenon in which alpha, the probability of making a Type 1 error, increases as you conduct more tests.
Adjustment of p-values	Accounting for doing multiple tests on the same data.
Overfitting	Fitting the noise.

2. Example – alpha inflation

Suppose there are 20 predictor variables and one outcome variable, all **randomly generated**. If 20 significance tests are done at $\alpha = 0.05$, the odds are good that at least one predictor will (falsely) turn out significant.

- Probability that one test correctly shows non-significant = 0.95.
- Probability that all 20 correctly show non-significant = $0.95 \times 0.95 \times 0.95 \ldots = 0.95^{20} =$ **0.36** (multiplication rule for independent events).
- Probability that at least one is falsely significant = $1 - 0.36 =$ **0.64**. This is **alpha inflation**.

$$\textbf{P(at least one false positive)} = 1 - (1 - \alpha)^n = 1 - 0.95^{20} = 0.64$$

3. Effect on statistical inference

- Spurious "significant" findings (Type 1 errors) appear, so conclusions are based on statistical noise.
- It is related to **overfitting** in data mining – "fitting the model to the noise". The more variables added or models run, the greater the chance something looks significant by chance.
- Results become hard to reproduce: **Bayer found in 2011 that when it tried to replicate 67 scientific studies, it could fully replicate only 14**; nearly two-thirds could not be replicated. More research does not necessarily mean better research.
- The number of journal articles nearly doubled between 2002 and 2010, giving many more opportunities to find "something interesting" by data dredging.

4. Common forms of multiplicity

- Checking for multiple pairwise differences across groups (e.g. A vs B, B vs C, A vs C).
- Looking at multiple subgroup results ("no significant treatment effect overall, but an effect for unmarried women younger than 30").
- Trying lots of statistical models and including lots of variables in models.
- Asking a number of different questions (different possible outcomes); looking at a clinical trial at multiple stages.

5. How it is corrected

(a) Adjustment procedures (for structured situations). They set the bar for significance more stringently than for a single test by "**dividing up the alpha**" according to the number of tests, giving a smaller alpha for each test.

- **Bonferroni adjustment:** simply divides alpha by the number of comparisons. Example: 20 tests at $\alpha = 0.05 \rightarrow$ each test is judged at $0.05/20 = 0.0025$.
- **Tukey's "honest significant difference" (Tukey's HSD):** used to compare multiple group means. It applies to the maximum difference among group means, comparing it to a benchmark based on the t-distribution (roughly equivalent to shuffling all the values together, dealing out resampled groups of the original sizes, and finding the maximum difference among the resampled group means).
- Dunnett's test is another procedure to adjust for multiple comparisons (further reading).

(b) Holdout sample and cross-validation (predictive modeling). In supervised learning, a **holdout set** where models are assessed on data they have not seen before mitigates the risk of an illusory model whose apparent efficacy is a product of chance. **Cross-validation** serves the same purpose.

(c) Resampling and simulation heuristics. For procedures without a labeled holdout set, rely on resampling and simulation to give random-chance benchmarks against which observed results can be compared.

(d) Awareness. The more you query and manipulate the data, the greater the role chance might play.

6. False discovery rate (FDR)

The term was originally used to describe the rate at which a given set of hypothesis tests would falsely identify a significant effect. It became particularly useful with **genomic research**, where massive numbers of tests are conducted during gene sequencing; researchers sought to set the testing process to control the FDR at a specified level. In data mining the term also means the misclassification rate within the class 1 predictions – the probability that a "discovery" (labelling a record as "1") is false.

7. Bottom line for data scientists

Situation	Remedy
Predictive modeling	Cross-validation and a holdout sample
Procedures without a labeled holdout set	Awareness that more querying means more chance + resampling/simulation chance benchmarks
Highly structured multiple comparisons	Statistical adjustment (Bonferroni, Tukey's HSD, Dunnett's test)

The adjustment procedures for highly defined and structured tests are too specific and inflexible to be of general use to data scientists.

Conclusion

Multiplicity increases the risk of concluding that something is significant just by chance (alpha inflation, overfitting, non-replicable results). It is controlled by adjusting alpha (Bonferroni, Tukey's HSD), by using holdout samples and cross-validation, and by comparing results against resampling-based chance benchmarks.

Q.6 b. What is the role of degrees of freedom in t – tests and other inferential statistics? [10 Marks]

1. Introduction – meaning of degrees of freedom

In the documentation and settings for many statistical tests and probability distributions you will see a reference to **"degrees of freedom" (d.f.)**. The concept applies to statistics calculated from sample data and refers to the **number of values free to vary**. The d.f. parameter, as applied to many probability distributions, affects the shape of the distribution.

Key term	Meaning
n or sample size	The number of observations (also called rows or records) in the data.
d.f.	Degrees of freedom.

Example: if you know the mean of a sample of 10 values, there are **9 degrees of freedom** – once 9 of the sample values are known, the 10th can be calculated and is not free to vary.

$$\textbf{d.f.} = n - 1$$

2. Role in variance and standard deviation (the n − 1 denominator)

Degrees of freedom is the name given to the **n − 1 denominator** seen in the calculations for variance and standard deviation. When a sample is used to estimate the variance of a population, using **n** in the denominator gives an estimate that is **slightly biased downward**. Using **n − 1** makes the estimate free of that bias.

3. Role in t-tests and other inferential statistics

- A large share of a traditional statistics course or text is consumed by standard tests of hypotheses (t-test, F-test, etc.).
- When sample statistics are standardized for use in traditional statistical formulas, **degrees of freedom is part of the standardization calculation**. It ensures that the standardized data matches the **appropriate reference distribution** (t-distribution, F-distribution, etc.).
- Thus d.f. is an **input** to many statistical tests and determines which t- (or F-) distribution the observed statistic is compared with, and hence the p-value.
- **Example from the t-test output:** `t = -1.0983, df = 27.693, p-value = 0.1408` – the d.f. identifies the t-distribution used to obtain the p-value. In the proportion test, `X-squared = 0.14893, df = 1` identifies the chi-square reference distribution.

4. Role in regression – factor (dummy) variables

There is one context in data science where d.f. is relevant: the use of **factored variables in regression** (including logistic regression). Some regression algorithms choke if exactly redundant predictor variables are present. This commonly occurs when factoring categorical variables into binary indicators (dummies).

- Consider "day of week". Although there are seven days, there are only **six degrees of freedom** in specifying the day. Once you know the day is not Monday to Saturday, it must be Sunday.
- Including the Mon–Sat indicators **and also Sunday** causes the regression to fail due to a **multicollinearity** error.
- Hence a factor with P levels is represented by **n − 1 (P − 1) indicator/dummy variables**; e.g. in the loan model the factor variables use reference coding with the reference levels left out (credit_card and MORTGAGE).

5. Importance for data science

- Is it important for data science? **Not really**, at least in the context of significance testing: formal statistical tests are used only **sparingly** in data science.

- The data size is usually large enough that it rarely makes a real difference whether the denominator has n or $n - 1$. As n gets large, the bias that would come from using n in the denominator **disappears**.
- Statisticians must navigate the many formula variations and learn their map; data scientists typically do not sweat the details of hypothesis tests the way a researcher preparing a paper might.

6. Summary table

Where d.f. appears	Role
Variance / standard deviation	$n - 1$ denominator removes the downward bias of the sample variance estimate
t-test, F-test and other tests	Part of standardizing the test statistic so it matches the correct reference distribution (t, F …)
Probability distributions	Controls the shape of the distribution
Regression with factor variables	$P - 1$ dummy variables are used to avoid multicollinearity

Key ideas

- The number of degrees of freedom forms part of the calculation to standardize test statistics so they can be compared to reference distributions (t-distribution, F-distribution, etc.).
- The concept lies behind the factoring of categorical variables into $n - 1$ indicator or dummy variables in regression (to avoid multicollinearity).

Conclusion

Degrees of freedom counts the number of values free to vary. It corrects the bias in variance estimation ($n - 1$), selects the correct reference distribution for t-tests and other tests, and explains why one dummy variable is dropped in regression. Its practical importance for data science is modest because of large data sizes.